Journal of the American Medical Informatics Association, 2025, **32(12)**, 1916–1926
https://doi.org/10.1093/jamia/ocaf138
Advance access publication 24 September 2025
Research and Applications

Research and Applications

A communication-efficient federated learning algorithm to assess racial disparities in post-transplantation survival time

Yudong Wang, PhD[1,2], Dazheng Zhang, PhD[1,2], Jiayi Tong, PhD[1,2,3], Xing He, PhD[4,5], Liang Li, PhD[6], Lichao Sun, PhD[7], Ashutosh M. Shukla, MD[8,9], Jiang Bian, PhD[4,5], David A. Asch, MD, MBA*[,10,11], Yong Chen, PhD*[,1,2,11]

[1]The Center for Health AI and Synthesis of Evidence (CHASE), Perelman School of Medicine, The University of Pennsylvania, Philadelphia, PA 19104, United States, [2]Department of Biostatistics, Epidemiology and Informatics, Perelman School of Medicine, The University of Pennsylvania, Philadelphia, PA 19104, United States, [3]Department of Biostatistics, Johns Hopkins Bloomberg School of Public Health, Baltimore, MD 21205, United States, [4]Department of Biostatistics and Health Data Science, Indiana University, Indianapolis, IN 46202, United States, [5]Regenstrief Institute, Indianapolis, IN 46202, United States, [6]Department of Biostatistics, The University of Texas MD Anderson Cancer Center, Houston, TX 77030, United States, [7]Department of Computer Science and Engineering, Lehigh University, Bethlehem, PA 18015, United States, [8]North Florida South Georgia Veteran Healthcare System, Gainesville, FL 32608, United States, [9]Division of Nephrology, Hypertension & Renal Transplantation, Department of Medicine, College of Medicine, University of Florida, Gainesville, FL 32610, United States, [10]Division of General Internal Medicine, The University of Pennsylvania, Philadelphia, PA 19104, United States, [11]Leonard Davis Institute of Health Economics, The University of Pennsylvania, Philadelphia, PA 19104, United States

*Corresponding authors: Yong Chen, PhD, Department of Biostatistics, Epidemiology, and Informatics, University of Pennsylvania, Blockley Hall 602, 423 Guardian Drive, Philadelphia, PA 19104, United States (ychen123@upenn.edu) and David A. Asch, MD, MBA, Division of General Internal Medicine, University of Pennsylvania, PCAM 6 South Tower—Dean's Suite, 3400 Civic Center Boulevard, Philadelphia, PA 19104, United States (asch@wharton.upenn.edu)

Y. Wang and D. Zhang contributed equally to this work.

Abstract

Objective: Patients of different race have different outcomes following renal transplantation. Patients of different race also undergo renal transplantation at different hospitals. We used a novel decentralized multisite approach to quantitatively assess the effect of site of care on racial disparities between non-Hispanic Black (NHB) and non-Hispanic White (NHW) patients in post-transplantation survival times.

Materials and Methods: In this study, we develop a communication-efficient federated learning algorithm to assess site-of-care associated racial disparities based on decentralized time-to-event data, called Communication-Efficient Distributed Analysis for Racial Disparity in Time-to-event Data (CEDAR-t2e). The algorithm includes 2 modules. Module I is to estimate the site-specific proportional hazards model for time-to-event outcomes in a distributed manner, in which the Poissonization is used to simplify the estimation procedure. Based on the estimated results from Module I, Module II calculates how long the kidney failure time of NHB patients would be extended had they been admitted to transplant centers in the same distribution as NHW patients were admitted.

Results: With application to United States Renal Data System data covering 39 043 patients across 73 transplant centers, we found no evidence suggesting the presence of site-of-care associated racial disparities in post-transplantation survival times. In particular, restricting to one year after transplantation, the counterfactual graft failure time would have been extended by only 0.61 days on average if NHB had the same admission distribution to transplant centers as NHW patients.

Discussion: The proposed approach offers a quantitative measure to evaluate site-of-care associated racial disparities.

Conclusion: Our approach has the potential to be extended to investigate site-of-care related disparities in other time-to-event outcomes, thus promoting health equity and improving patient health in various fields.

Key words: communication-efficient; federated learning; kidney transplantation; racial disparity; time-to-event outcomes.

Introduction

Kidney transplantation is the best approach to manage end-stage renal disease (ESRD).[1–5] Non-Hispanic Black (NHB) patients are less likely to receive kidney transplants, experience longer waiting times, exhibit higher graft failure rates, and achieve shortened survival times[6–10] when compared with non-Hispanic White (NHW) patients with similar demographic (eg, age and sex), clinical (eg, comorbidities), and socioeconomic characteristics (eg, income level and insurance status). Site of care may play an important role in driving these disparities.[11] A long history of racial residential segregation locates different racial groups next to different transplant centers with varying quality of site,[12,13] which may lead to disparities in post-transplantation survival times.

Post-transplantation survival time, also known as the kidney graft failure time, is a critical time-to-event outcome that

Received: October 9, 2024; **Revised:** June 15, 2025; **Editorial Decision:** July 20, 2025; **Accepted:** August 1, 2025
© The Author(s) 2025. Published by Oxford University Press on behalf of the American Medical Informatics Association. All rights reserved.
For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site—for further information please contact journals.permissions@oup.com.

Downloaded from https://academic.oup.com/jamia/article/32/12/1916/8262716 by guest on 05 March 2026

measures the time of graft failure or patient death following a transplant procedure. As the patients are not randomly allocated to different hospitals, direct observational comparisons of how transplant hospitals differ in time-to-event outcomes may be confounded by case-mix variation[14] and residential segregation which leads patients of one race to disproportionately get care at some hospitals rather than others.[12,13]

To evaluate the potential site-of-care associated racial disparity, the most relevant study is Asch et al[6] which assessed the association between site of care and racial disparities in mortality rates of patients hospitalized with COVID-19 infection. In Asch et al, a counterfactual modeling approach was used to adjust for patient-level characteristics and the method estimated the mortality rates for NHB patients had they been admitted to the hospitals where NHW patients were admitted instead. Although Asch et al provided an effective approach to assess the effect of site of care, it was restricted to binary outcomes rather than time-to-event outcomes like post-transplantation survival times. It also relied on a single centralized data source of multisite insurance claims data.

Over the past decades, the significant expansion of distributed research networks (DRNs) within healthcare systems has provided new opportunities to utilize real-world data (RWD) to assess the impact of care location on time-to-event outcomes. Despite these opportunities, the DRNs—especially decentralized ones (eg, OHDSI,[15] STAR Clinical Research Network,[16] The 4CE[17] Consortium) where patient-level data cannot be shared across sites[15,18]—present practical challenges in analyzing RWD. First, conducting federated learning without requiring patient-level but rather using aggregated data presents a primary challenge, given regulatory concerns and proprietary business interests. Many methods have been developed to tackle this challenge under different models,[18–20] but none of them considered the context of racial disparity for time-to-event outcomes. Additionally, significant variations in patient characteristics can confound analyses,[14] thereby obscuring the true effects of hospital-level care.[21]

To address these challenges, we developed a federated learning framework that can assess the site-of-care associated racial disparities in time-to-event outcomes based on decentralized data. Our framework is called Communication-Efficient Distributed Analysis for Racial Ddisparity in Ttime-to-Event Data (CEDAR-t2e). It consists of 2 modules. The first is a federated learning module for statistical models of time-to-event outcomes. The second is to predict the counterfactual graft failure time of NHB patients if they were admitted to transplant centers in the same distribution as NHW patients. The difference between the counterfactual and the actual graft failure times of NHB patients is then used to measure how long the failure times of NHB patients would be extended if they had the same access to the transplant center as NHW patients. This approach enables us to evaluate the racial disparities associated with site-of-care. Our model offers several benefits:

1) It addresses the data-sharing barrier in that it only requires each site to share some summary-level statistics.
2) It requires only 3 rounds of communication across sites.
3) It provides numerically stable results for assessing site-of-care-associated racial disparities.
4) It assesses the site-of-care-associated racial disparities through counterfactual modeling.

The number of communications of our algorithm is significantly smaller than that for some iterative federated algorithms such as GLORE[22] for logistic regression, dPQL[23] for generalized linear mixed-effects model (GLMM), and Fed-GLMM[24] for GLMMs with longitudinal and correlated measurements. In those iterative federated algorithms, the total number of communication rounds equals the total number of iterations required to achieve convergence, which could be dozens or even hundreds. Our algorithm requires only 3 rounds of communications, which is significantly smaller than those approaches. In particular, 3 rounds of communications can be handled manually across data partners, for example, via secure emails or web-based secure data sharing platform, PDA-OTA.[25] It avoids the need for demanding infrastructure for automated data exchange of aggregated data across all data partners, which is typically needed for iterative algorithms[22–24] requiring numerous rounds of communications. As a result, our approach is *communication-efficient* and it can be conveniently deployed in real-world decentralized applications.

To demonstrate the performance and applicability of the proposed CEDAR-t2e algorithm, we applied it to a real-world dataset obtained from the United States Renal Data System (USRDS)[26] that collects and analyzes information about chronic kidney disease and ESRD in the US. Though CEDAR-t2e was originally developed for evaluating racial disparities in post-transplantation survival times, it is actually a generic methodological framework that can evaluate site-of-care-related disparities in general time-to-event outcomes. It has the potential to be applied in various areas to advance health equity and improve health outcomes.

Materials and methods

Notation

Suppose there are totally J clinical sites, and the jth site includes n_j recipients, $j = 1, \ldots, J$. Let T_{ij} be the post-transplantation survival time, that is, the time to graft failure or death, of the ith recipient in the jth clinical site, and C_{ij} be the random censoring time, which is the duration from the time when the recipient receives transplantation to the time when study ends $(i = 1, \ldots, n_j)$. Denote $X_{ij} = \min(T_{ij}, C_{ij})$ and $\delta_{ij} = I\{T_{ij} \le C_{ij}\}$ as the censoring indicator, where $\delta_{ij} = 1$ indicates the ith recipient in the jth site had the kidney graft failure or death before the end of study, and $\delta_{ij} = 0$ implies the recipient is censored at the end of study. In addition, let Z_{ij} be the vector of covariates for the ith recipient in the jth site including race as well as other characteristics such as age and gender. The observed data in the jth site are $D_j = \{X_{ij}, \delta_{ij}, Z_{ij}\}_{i=1}^{n_j}, j = 1, \ldots, J$.

We consider the following proportional hazards model for the hazard function $\lambda_{ij}(t)$ of T_{ij}:

$$\lambda_{ij}(t) = \lambda_{0j}(t)\exp\{\boldsymbol{\beta}^\top Z_{ij}\}. \quad (1)$$

In this model, we assume that the regression parameters $\boldsymbol{\beta}$ for patient-level covariates Z_{ij} are shared across all sites, implying that the effects of covariates such as age, gender, and comorbidities are consistent and do not vary by geographic location. This reflects the belief that these covariate effects are largely intrinsic and stable across different populations. In contrast, we allow the baseline hazard functions $\lambda_{0j}(t)$ to be site-specific, thereby accommodating cross-site heterogeneity in survival distributions. For instance, patients in

Downloaded from https://academic.oup.com/jamia/article/32/12/1916/8262716 by guest on 05 March 2026

certain sites may systematically experience longer post-transplantation survival times due to access to better healthcare infrastructure, while others may have shorter survival times. Allowing site-specific baseline hazards provides the flexibility to model such differences in underlying risk. This modeling framework—assuming common regression parameters while allowing site-specific baseline hazards—is widely adopted in the literature on federated Cox regression models.[27,28] Following the literature of survival analysis,[29,30] we model the logarithmic baseline hazard function by spline functions:

$$\log\lambda_{0j}(t) = a_j + \boldsymbol{b}^\top B_j(t),$$

where a_j is the intercept term for the logarithm baseline hazard and it represents the overall risk of graft failure for patients in site j—a larger value of a_j indicates greater risk for recipients in site j. To account for the heterogeneity in overall risk across transplant centers, we allow a_j to come from a normal distribution $N(\mu_a, \sigma_a^2)$, where the mean parameter μ_a represents the average level of the overall risks in different centers and the variance represents the variability of the overall risks across centers. In addition, $B_j(t) = [B_{j1}(t), B_{j2}(t), \ldots, B_{jK}(t)]$ is a vector of K known spline basis functions, and $b = (b_1, b_2, \ldots, b_K)$ is the vector of spline coefficients. There are many potential choices for the spline basis functions $B_j(t) = [B_{j1}(t), B_{j2}(t), \ldots, B_{jK}(t)]$, such as the linear bases[31,32] and cubic bases,[29,30] which lead to linear splines and cubic splines, respectively. Basically, linear splines use piecewise linear functions to approximate the target function (ie, the logarithmic baseline hazard in our context), while the cubic splines use piecewise cubic polynomials for approximation, both of which can approximate the target function arbitrarily well as the number of bases (ie, K) grows to infinity. Compared to linear splines, cubic splines are more flexible and lead to smooth estimates for the logarithmic baseline hazard. However, there is also risk of overfitting for cubic splines due to the increased flexibility and its computational cost is higher than that for linear splines. To avoid overfitting and achieve higher computational efficiency, we choose the linear spline basis functions as follows:

$$B_{jk}(t) = (t - t_{jk})_+$$

where $0 < t_{j1} < t_{j2} < \cdots < t_{jK}$ are predetermined knots.

Statistical estimation using Poissonization for pooled time-to-event data

We first discuss the estimation of the random-effects proportional hazards model when the patient-level data can be pooled together. Before presenting the details, we clarify 2 key distinctions between our model compared to one recent study on multisite Cox proportional hazard models, that is, the multisite Cox model with time-varying coefficients in Liang et al.[33] First, Liang et al focus on estimating the time-varying regression coefficients only, treating the baseline hazard functions as nuisance parameters which are not estimated. Consequently, their estimation procedure is based on a kernel-weighted version of the standard partial likelihood. In contrast, our study targets the joint estimation of both the regression coefficients (which is time-invarying in our model) and the baseline hazard functions, which requires us to work with the full likelihood rather than the partial likelihood. Second, Liang et al assume that the baseline hazards in different sites are fixed, whereas we allow them to be stochastic by introducing random intercepts a_j. Because these random intercepts a_j are unknown, we need to integrate them out, leading to a complex integral with respect to the intercept in the full likelihood. Without additional tools, direct maximization of the full likelihood would be numerically challenging. To address this challenge, we perform Poissonization to transform the proportional hazards model in eqn (1) to an equivalent Poisson regression model. This transformation facilitates tractable computation and enables scalable estimation.

Let U be the maximum follow-up duration and $0 = \tau_{-1} = \tau_0 < \tau_1 < \cdots < \tau_D = U$ be a sequence of grid points over the whole follow-up period $[0, U]$. Given i and j, let M_{ij} be the index such that $\tau_{M_{ij}-1} < X_{ij} \le \tau_{M_{ij}}$. Define $Y_{ijg} = 0$ for $g = 0, 1, \ldots, M_{ij} - 1$ and $Y_{ijg} = \delta_{ij}$ for $g = M_{ij}$. For site j, $j = 1, 2, \ldots, J$, the proportional hazards model in eqn (1) is approximately equivalent to the following Poisson regression model:

$$\log\big(E[Y_{ijg} \mid Z_{ij}]\big) = a_j + \boldsymbol{\beta}^\top Z_{ij} + \boldsymbol{b}^\top B_j\big(\min\{X_{ij}, \tau_g\}\big) + \log c_{ijg}, \tag{2}$$

where $c_{ijg} = \big(min\{X_{ij}, \tau_{g+1}\} - \tau_{g-1}\big)/2$ is the offset value, $g = 0, 1, \ldots, M_{ij}$, $i = 1, 2, \ldots, n_j$. It can be shown that as the grid points become increasingly dense over $[0, U]$, the log-likelihood function of the Poisson regression model converges to that of the original proportional hazards model. This suggests that the information loss due to Poissonization becomes negligible when the grid points are sufficiently dense. For our analysis, we use 20 equally spaced grid points over $[0, U]$, which is dense enough to avoid significant information loss.

The Poisson regression model in eqn (2) is a standard GLMM, which is much easier to estimate than the proportional hazards model in eqn (1) and it allows us to model the patient-level and hospital-level characteristics at the same time. When patient-level data can be shared across sites, the pooled estimator of the fixed effects $(\boldsymbol{\beta}, \boldsymbol{b})$ and random effects $(a_1, a_2, \ldots, a_J)$ can be readily obtained by fitting the GLMM using the maximum likelihood approach. The estimated fixed and random effects can then be used to predict the graft failure time for each patient.

Proposed method for distributed data: CEDAR-t2e

In practice, the sharing of patient-level data across sites has become one of the key challenges due to various data protection and regulation requirements within the research networks, especially in international collaborations. To address this, it is crucial to develop federated learning approaches that only require aggregated data to achieve the same analysis task as we have the pooled patient-level data.

In response, we develop a federated learning framework called CEDAR-t2e to evaluate the site-of-care associated racial disparities in post-transplantation survival times. This framework is designed for the proposed heterogeneous Cox model when the patient-level data cannot be pooled. It consists of 2 major modules as follows.

Downloaded from https://academic.oup.com/jamia/article/32/12/1916/8262716 by guest on 05 March 2026

Module I: federated learning of the model for time-to-event outcomes

Step (a): Each local site converts time-to-event data into GLMM-format, fits the local GLMM, and transfers the local estimators $(\widehat{\boldsymbol{\beta}}_j, \widehat{\boldsymbol{b}}_j)$ of fixed effects $(\boldsymbol{\beta}, \boldsymbol{b})$ to the coordinating center.

Step (b): Coordinating center estimates the overall fixed effects $(\widehat{\boldsymbol{\beta}}, \widehat{\boldsymbol{b}})$ via meta-analysis and broadcasts the overall fixed effects $(\widehat{\boldsymbol{\beta}}, \widehat{\boldsymbol{b}})$ to all local sites, after which each local site estimates the local random effect $\widehat{a}_j$ and transfers it back to the coordinating center.

Step (c): Coordinating center calibrates the local random effects $(\widehat{a}_1, \widehat{a}_2, \ldots, \widehat{a}_J)$ via meta-regression and broadcasts the calibrated random effects $(\tilde{a}_1, \tilde{a}_2, \ldots, \tilde{a}_J)$ to all local sites.

Module II: counterfactual modeling of graft failure time

Step (d): Coordinating center calculates the admission distribution $\boldsymbol{p} = (p_1, p_2, \ldots, p_J)$ transplant centers for NHW patients and broadcast it to all local sites.

Step (e): Each local site calculates the factual/counterfactual post-transplantation survival times for NHB patients, in which "counterfactual" means we assume NHB patients were admitted to transplant center in the same admission distribution $\boldsymbol{p} = (p_1, p_2, \ldots, p_J)$ as NHW patients. The calculated factual/counterfactual survival times are then transferred to the coordinating center, and the coordinating center combines the results from all local sites to obtain a final measure for racial disparities.

The workflow of CEDAR-t2e is displayed in Figure 1 and more details on the formulation and implementation of the 2 modules are available in the following two sections.

Module I: federated learning of the model for time-to-event outcomes

In Module I, we develop a federated learning algorithm to estimate the parameters in the random-effects proportional hazards model based on decentralized data. This algorithm comprises 3 main steps as detailed below.

First, as mentioned earlier, it is numerically unstable to directly estimate the random-effects proportional hazards model, especially in a federated manner. An effective alternative is to estimate the equivalent GLMM, that is, the Poisson regression model in eqn (2). Therefore, in the first step of Module I, each site locally converts the original time-to-event data $\{(X_{ij}, \delta_{ij}, Z_{ij}); i = 1, 2, \ldots, n_j\}$ into the GLMM format $\{(X_{ij}, Y_{ijg}, c_{ijg}, Z_{ij}); i = 1, \ldots, n_j, g = 0, 1, \ldots, M_{ij}\}$. Based on the GLMM-format data, each site locally fits the GLMM in eqn (2) using the maximum likelihood approach, obtaining the local estimators $\widehat{\boldsymbol{\beta}}_j, \widehat{\boldsymbol{b}}_j$ for fixed effects $\boldsymbol{\beta}, \boldsymbol{b}$ and the respective variance estimators $V_{\beta,j}, V_{b,j}$ of $\widehat{\boldsymbol{\beta}}_j, \widehat{\boldsymbol{b}}_j$. These local estimators $\widehat{\boldsymbol{\beta}}_j, \widehat{\boldsymbol{b}}_j, V_{\beta,j}, V_{b,j}$ are then transferred to the coordinating center.

Second, the coordinating center integrates the local estimators of fixed effects $\widehat{\boldsymbol{\beta}}_j, \widehat{\boldsymbol{b}}_j$, $j = 1, \ldots, J$, via meta-analysis to improve the estimation efficiency. More specifically, the coordinating site calculates the inverse-variance-weighted estimator for fixed effects $\boldsymbol{\beta}$ and $\boldsymbol{b}$:

$$\widehat{\boldsymbol{\beta}} = \left[\sum_{j=1}^{J} V_{\beta,j}^{-1}\right]^{-1} \sum_{j=1}^{J} V_{\beta,j}^{-1} \widehat{\boldsymbol{\beta}}_j, \quad \widehat{\boldsymbol{b}} = \left[\sum_{j=1}^{J} V_{b,j}^{-1}\right]^{-1} \sum_{j=1}^{J} V_{b,j}^{-1} \widehat{\boldsymbol{b}}_j.$$

The calculated overall fixed effects $(\widehat{\boldsymbol{\beta}}, \widehat{\boldsymbol{b}})$ are then broadcasted to all local sites. Next, each local site j plugs the overall fixed effects $(\widehat{\boldsymbol{\beta}}, \widehat{\boldsymbol{b}})$ into eqn (2) and refit the plug-in model using maximum likelihood approach, obtaining the estimator

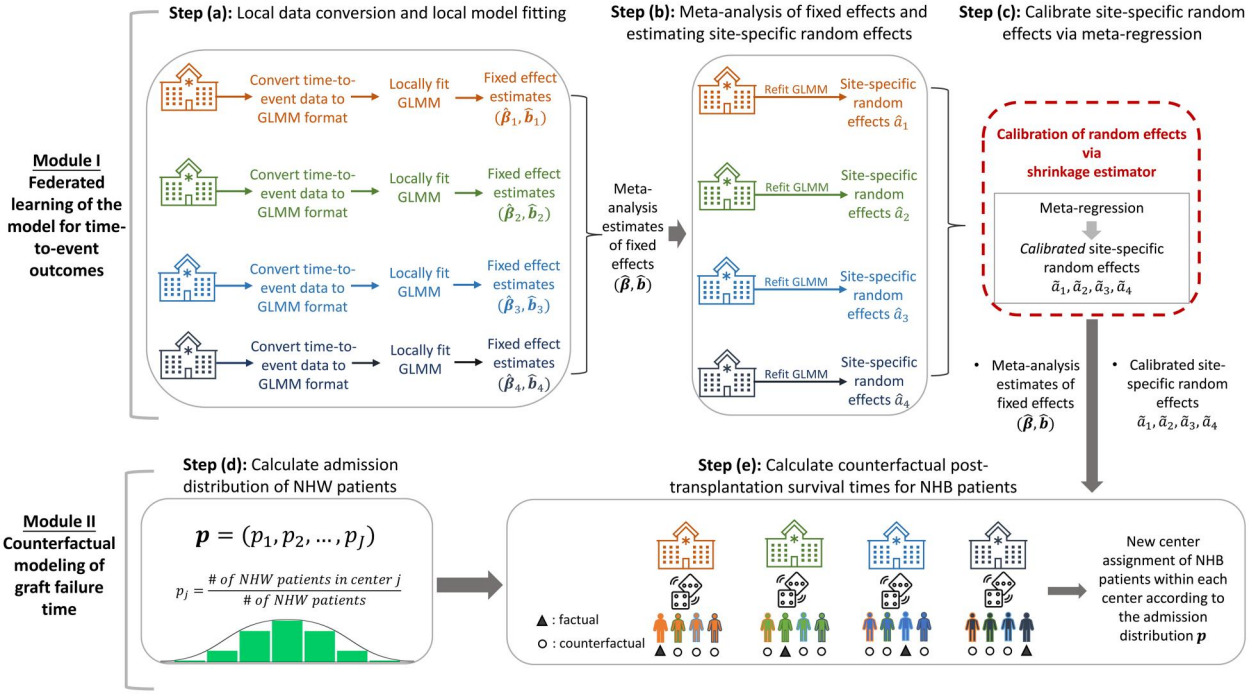


Figure 1. The working flow of CEDAR-t2e. Module I is to estimate parameters of the time-to-event model in a federated manner. Module II is to calculate the counterfactual post-transplantation survival time for NHB patients assuming that they had the same admission distribution to transplant centers as NHW patients.

Downloaded from https://academic.oup.com/jamia/article/32/12/1916/8262716 by guest on 05 March 2026

$\widehat{a}_j$ of the site-specific random effect a_j and the respective variance estimator ξ_j^2 of $\widehat{a}_j$. The estimators $(\widehat{a}_j, \xi_j^2)$, together with some site-level covariate Y_j, are then transferred to the coordinating center.

Third, the coordinating center calibrates the estimated random effects $\widehat{a}_1, \ldots, \widehat{a}_J$ through meta-regression.[34] Following the idea of the James-Stein estimator,[35] the coordinating center shrinks the estimators of site-specific random effects $\widehat{a}_1, \ldots, \widehat{a}_J$ via the following regression model:

$$\widehat{a}_j = u_j + \eta^\top Y_j + \varepsilon_j,$$

where $u_j \sim N(\mu_u, \sigma_u^2)$, Y_j is the site-level covariate, and $\varepsilon_j \sim N(0, \xi_j^2)$ is the error term. Specifically, the calibrated site-specific random effect is calculated as $\tilde{a}_j = \widehat{u}_j + \widehat{\eta}^\top Y_j$, $j = 1, \ldots, J$, where $\widehat{u}_j$ and $\widehat{\eta}$ are the maximum likelihood estimators. These calibrated random effects are then broadcasted to all local sites.

The above 3 steps take 2 rounds of communication between coordinating center and local sites, with one round defined as a complete exchange of information from the local site to the coordinating center and back. After those steps, every local site obtains the overall fixed effects $(\widehat{\boldsymbol{\beta}}, \widehat{\boldsymbol{b}})$ and the calibrated random effects $\tilde{a}_1, \tilde{a}_2, \ldots, \tilde{a}_J$. These estimators will be used to assess the site-of-care associated disparities in the next module.

Module II: counterfactual modeling

Module II adopts the counterfactual modeling approach to account for racial disparities associated with the site of care. In practice, NHB and NHW patients exhibit different patterns when selecting and gaining admission to transplant centers, leading to different admission distributions for the 2 groups. To address the question of whether NHB patients would experience better post-transplantation outcomes if these access differences did not exist, we make the counterfactual assumption that NHB patients would have the same admission distribution as NHW patients. In other words, we assume that NHB patients are choosing and being admitted to transplant centers as if they were NHW. This is equivalent to assuming that NHB and NHW patients had the same patterns in the factors that influence the selection of transplant centers, such as socioeconomic status and access to better facilities. This counterfactual assumption allows us to estimate how post-transplantation survival times for NHB patients might change if they had the same patterns in selecting and being admitted to transplant centers as NHW patients. It is central to evaluating whether site-of-care disparities contribute significantly to racial disparities in post-transplantation survival outcomes.

This module simulates the time-to-event outcome, that is, post-transplantation survival time, for NHB patients under the counterfactual case that NHB patients received their care in transplant centers in the same distribution as NHW patients received their care. The hypothesis is that the counterfactual post-transplantation survival time would be longer than the observed post-transplantation survival time for NHB patients if site-of-care-associated racial disparities exist for NHB patients receiving kidney transplants.

The workflow of Module II is illustrated in Figure 2. First, each local site transfers the number of NHW and NHB patients to the coordinating center, and the coordinating center calculates the admission distribution of NHW patients. Let n_j^W and n_j^B be the number of NHW and NHB patients in the jth site, respectively. The coordinating center calculates the admission distribution $\boldsymbol{p} = (p_1, p_2, \ldots, p_J)$ of NHW patients over the J sites, where $p_j = n_j^W / \sum_{j=1}^J n_j^W$. Then, the coordinating center broadcasts the admission distribution $\boldsymbol{p}$ as well as the estimated random effects $(\tilde{a}_1, \tilde{a}_2, \ldots, \tilde{a}_J)$ to every local site. The cross-site communication in this step is implemented together with the second rounds of communication in Module I. More specifically, the transfer of (n_j^W, n_j^B) from local sites to the coordinating center can be done when site j transfers the site-specific random effects $(\widehat{a}_j, \xi_j^2)$ to the coordinating center in the second step of Module I; the broadcast of admission distribution $\boldsymbol{p}$ can be done together with the broadcast of $(\tilde{a}_1, \tilde{a}_2, \ldots, \tilde{a}_J)$ in the third step of Module I. Thus, we do not incur any additional cross-site communication in this step.

Second, each local site calculates the counterfactual post-transplantation survival times for NHB patients. We use site j to illustrate this procedure. Given a NHB patient in site j, for example, the ith NHB patient in site j, we randomly assign this patient to one hypothetical transplant center j^* based on the admission distribution $\boldsymbol{p}$ of NHW patients. Then, based on the estimators $\widehat{\boldsymbol{\beta}}$, $\widehat{\boldsymbol{b}}$, $(\tilde{a}_1, \tilde{a}_2, \ldots, \tilde{a}_J)$ obtained from Module I, we calculate the counterfactual restricted mean survival time (RMST)[36–38] for the given NHB patient as if they got transplantation in center j^*. Specifically, the counterfactual RMST within a predetermined window $[0, t]$ is calculated as

$$\mathrm{RMST}_{j,j^*}(t; Z_{ij}) = \int_0^t \exp\{-\exp\{\widehat{\boldsymbol{\beta}}^\top Z_{ij}\}\widehat{\Lambda}_{j^*}(s)\}ds,$$

where $\widehat{\Lambda}_{j^*}(s) = \int_0^s \exp\{\tilde{a}_{j^*} + \widehat{\boldsymbol{b}}^\top \boldsymbol{B}(u)\}du$ is the estimated cumulative hazard function in the hypothetical center j^*. If $j^* = j$, then $\mathrm{RMST}_{j,j}(t; Z_{ij})$ is the factual RMST that measures the expected graft failure time of the ith NHB patient in site j. Both the counterfactual and factual RMSTs are calculated for every NHB patient in site j. Denote the average of the counterfactual RMSTs of NHB patients in site j as $\mathrm{Counter_RMST}_j(t)$. Similarly, $\mathrm{Fact_RMST}_j(t)$ is the average of the factual RMSTs of NHB patients in site j. Then, the difference $\mathrm{Counter_RMST}_j(t) - \mathrm{Fact_RMST}_j(t)$ is used to measure how long the post-transplantation survival time of NHB patients in site j would be extended on average if they had the same access to transplant centers as NHW patients.

The values of $\mathrm{Counter_RMST}_j(t)$ and $\mathrm{Fact_RMST}_j(t)$ will be transferred to the coordinating center, and the coordinating center then integrates the information from all sites. Specifically, the coordinating center calculates the weighted average of counterfactual RMSTs over all sites, obtaining an overall counterfactual RMST for all NHB patients across J sites:

$$\mathrm{Counter_RMST}(t) = \frac{\sum_{j=1}^J n_j^B \mathrm{Counter_RMST}_j(t)}{\sum_{j=1}^J n_j^B}.$$

Similarly, we can also obtain an overall factual RMST for all NHB patients across J sites:

$$\mathrm{Fact_RMST}(t) = \frac{\sum_{j=1}^J n_j^B \mathrm{Fact_RMST}_j(t)}{\sum_{j=1}^J n_j^B}.$$

The difference between counterfactual and factual RMSTs, that is, $\mathrm{Counter_RMST}(t) - \mathrm{Fact_RMST}(t)$, quantifies how

Downloaded from https://academic.oup.com/jamia/article/32/12/1916/8262716 by guest on 05 March 2026

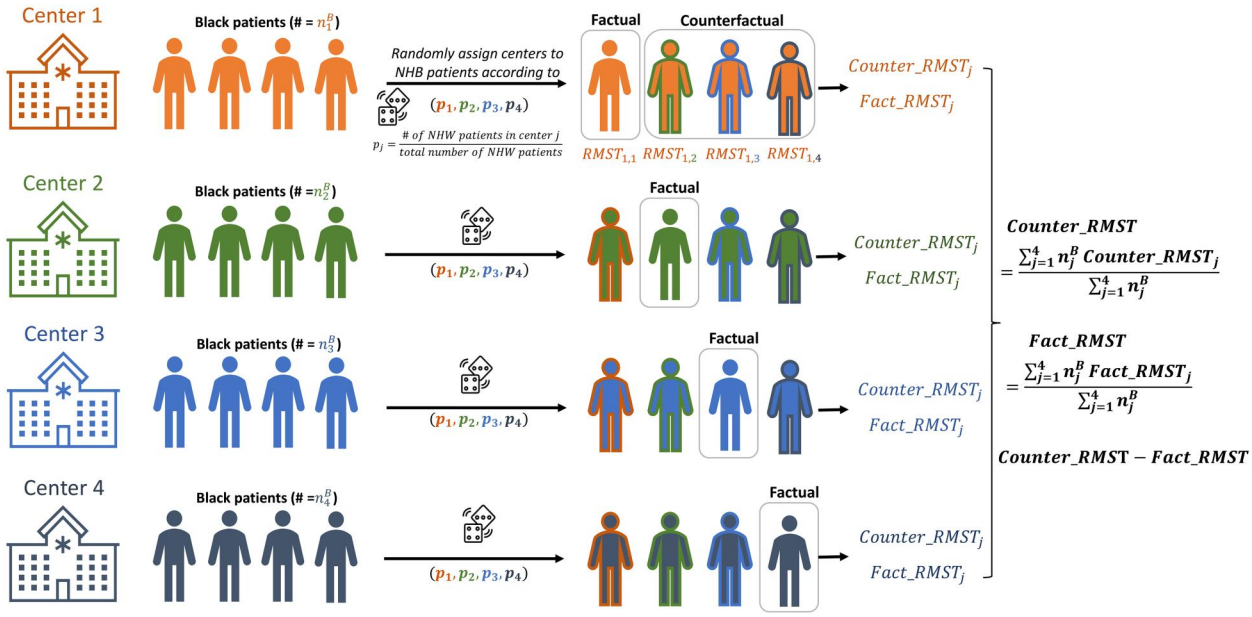


Figure 2. The working flow of Module II. Each NHB patient is randomly assigned to a counterfactual transplant center according to the distribution $p = (p_1, p_2, \ldots, p_J)$ that NHW patients were admitted. The counterfactual RMST is calculated and compared with the factual RMST, and their difference can be used to measure the site-of-care associated racial disparities in post-transplantation survival times.

long the graft failure time of a NHB patient would be extended on average if they had the same admission distribution as NHW patients were admitted. This difference is expected to be positive if there exist site-of-care-associated racial disparities for NHB patients receiving kidney transplants.

To account for simulation uncertainties in generating the hypothetical centers for NHB patients, we repeat the above calculation of Counter_RMST(t) L times, and we denote the values of the rth replication as Counter_RMST$^{(r)}(t)$, $r = 1, 2, \ldots, L$. The CIs of Counter_RMST(t) and Counter_RMST(t) − Fact_RMST(t) are then obtained based on the L replications.

In this counterfactual modeling approach, different centers only need to share the estimates $\tilde{a}_j$ of random effects and the total number of NHB and NHW patients. This shows the advantage of our method that it does not require sharing patient-level data across centers. In addition, although we repeat the calculation of counterfactual RMSTs for L times, there only requires one round of cross-site communication in this module. Each site can implement the L replications locally and then transfer all L replications to the coordinating center all at once. Therefore, this module is also communication efficient.

Detailed procedures of Modules I and II are summarized in Algorithm S1 in the Supplementary Materials.

Results

We used the proposed CEDAR-t2e framework to evaluate whether there exist site-of-care-associated racial disparities for NHB patients in survival times after kidney transplantation. More specifically, based on a real registry dataset from 73 transplant centers, we quantified how long the post-transplantation survival time for NHB patients would be extended had they been admitted to transplant centers in the same distribution as NHW patients selected the transplant centers.

The dataset that we used was collected from the USRDS. It covered the population of NHB and NHW adults who underwent kidney transplantation between January 1, 2009, and December 31, 2018. Assessments of data quality were performed at each center before analysis, including checking for missing values and an adequate number of cases. In particular, we included all transplant centers with sample size no less than 300, and centers failing to meet the quality criteria were removed from the analysis. This resulted in a final analysis cohort comprising 39 043 patients across 73 transplant centers. The number of patients in each center varied from 303 to 1097. Among all patients, 16 688 were NHB (42.7%), and 22 355 were NHW (57.3%); 3037 patients (7.9%) experienced either death or graft failure within the one-year period after transplantation. For each patient, there were a number of available patient-level covariates, including recipient race, age, gender, BMI, diabetes status, cold ischemia time, and total ESRD time at transplant, as well as donor race, age, gender, BMI, hypertension status, and history of comorbidities such as AIDS, alcohol dependence, COPD, peripheral vascular disease, and diabetes. In our analysis, we selected 17 patient-level covariates based on statistical significance and clinical importance suggested by physicians, as summarized in Table 1. In addition to patient-level covariates, we also included one center-level covariate, that is, the number of transplants from each transplant center.

We first present the proportions of admitted NHB patients across 73 transplant centers in Figure 3. It can be seen the percentages of NHB patients showed significant variation across centers. The center with the highest NHB proportion had a percentage as high as >80%, while those with low NHB proportions only had less than 20% NHB patients, indicating that NHB and NHW patients were distributed unevenly across centers. Such heterogeneity in racial

Downloaded from https://academic.oup.com/jamia/article/32/12/1916/8262716 by guest on 05 March 2026

Table 1. Summary of patient-level characteristics in the kidney transplant data.

Characteristics	Total (N = 39 043)	Black (N = 16 688)	White (N = 22 355)
Recipient race			
Black	16 688 (42.74%)	16 688 (100%)	0 (0%)
White	22 355 (57.26%)	0 (0%)	22 355 (100%)
Recipient gender			
Male	24 082 (61.68%)	9971 (59.75%)	14 111 (63.12%)
Female	14 961 (38.32%)	6717 (40.25%)	8244 (36.88%)
Recipient age at transplant			
18-25	854 (2.19%)	285 (1.71%)	569 (2.55%)
35-45	5960 (15.27%)	3075 (18.43%)	2885 (12.91%)
45-55	9688 (24.81%)	4493 (26.92%)	5195 (23.24%)
55-65	11 890 (30.45%)	4905 (29.39%)	6985 (31.25%)
65-75	6474 (16.58%)	2253 (13.5%)	4221 (18.88%)
≥75	835 (2.14%)	214 (1.28%)	621 (2.78%)
Recipient body mass index			
Under	652 (1.67%)	279 (1.67%)	373 (1.67%)
Over	13 046 (33.41%)	5331 (31.95%)	7715 (34.51%)
Obesity	15 592 (39.94%)	7110 (42.61%)	8482 (37.94%)
Recipient total cold ischemia time			
>20 hours	11 143 (28.54%)	5322 (31.89%)	5821 (26.04%)
⩽20 hours	27 900 (71.46%)	11 366 (68.11%)	16 534 (73.96%)
Donor race			
Black	7170 (18.36%)	4815 (28.85%)	2355 (10.53%)
Asian	757 (1.94%)	311 (1.86%)	446 (2%)
Donor gender			
Male	21 644 (55.44%)	9542 (57.18%)	12 102 (54.14%)
Female	17 399 (44.56%)	7146 (42.82%)	10 253 (45.86%)
Donor age			
<10	1317 (3.37%)	613 (3.67%)	704 (3.15%)
10-20	2411 (6.18%)	1168 (7%)	1243 (5.56%)
30-40	7147 (18.31%)	3056 (18.31%)	4091 (18.3%)
40-50	8852 (22.67%)	3792 (22.72%)	5060 (22.63%)
50-60	8819 (22.59%)	3627 (21.73%)	5192 (23.23%)
≥60	3637 (9.32%)	1362 (8.16%)	2275 (10.18%)
Donor body mass index			
Under	1898 (4.86%)	858 (5.14%)	1040 (4.65%)
Over	12 781 (32.74%)	5174 (31%)	7607 (34.03%)
Obesity	11 864 (30.39%)	5308 (31.81%)	6556 (29.33%)
Donor history of hypertension			
Yes	9989 (25.58%)	4643 (27.82%)	5346 (23.91%)
No	29 054 (74.42%)	12 045 (72.18%)	17 009 (76.09%)
Duration of ESRD			
1-5 years	19 657 (50.35%)	7206 (43.18%)	12 451 (55.7%)
≥5 years	16 198 (41.49%)	8920 (53.45%)	7278 (32.56%)
Comorbidities			
Alcohol dependence			
Yes	343 (0.88%)	147 (0.88%)	196 (0.88%)
No	38 700 (99.12%)	16 541 (99.12%)	22 159 (99.12%)
Chronic obstructive pulmonary disease			
Yes	656 (1.68%)	224 (1.34%)	432 (1.93%)
No	38 387 (98.32%)	16 464 (98.66%)	21 923 (98.07%)
Diabetes (currently on insulin)			
Yes	8849 (22.66%)	3717 (22.27%)	5132 (22.96%)
No	30 194 (77.34%)	12 971 (77.73%)	17 223 (77.04%)
Peripheral vascular disease			
Yes	1525 (3.91%)	504 (3.02%)	1021 (4.57%)
No	37 518 (96.09%)	16 184 (96.98%)	21 334 (95.43%)
Diabetes (primary or contributing)			
Yes	1153 (2.95%)	721 (4.32%)	432 (1.93%)
No	37 890 (97.05%)	15 967 (95.68%)	21 923 (98.07%)
AIDS			
Yes	42 (0.11%)	** (**)	** (**)
No	39 001 (99.89%)	** (**)	** (**)
Graft failure/death	3073 (7.87%)	1417 (8.49%)	1656 (7.41%)

Some data values are masked (denoted by **) to protect patient privacy.

Downloaded from https://academic.oup.com/jamia/article/32/12/1916/8262716 by guest on 05 March 2026

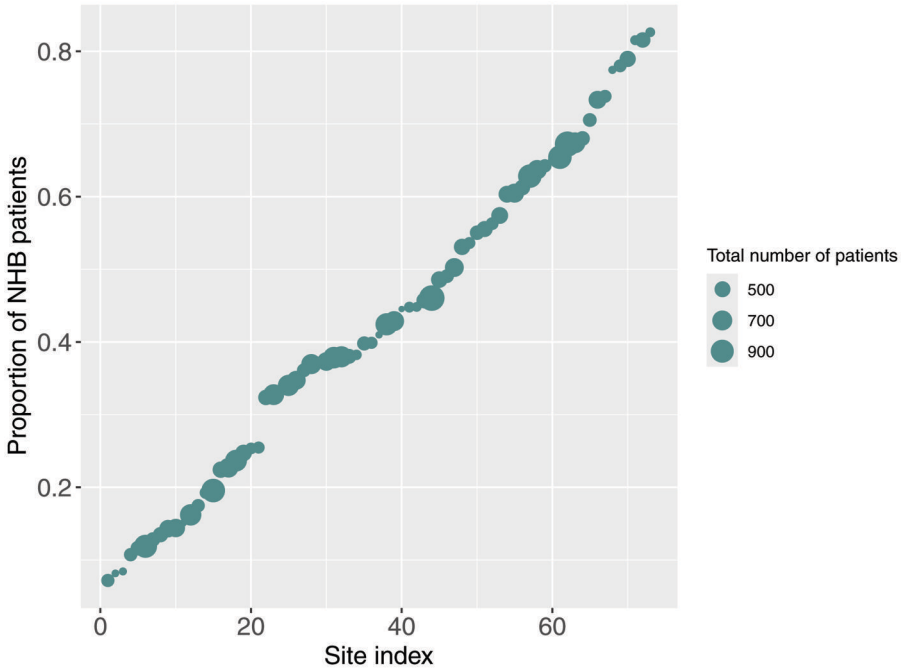


Figure 3. Proportions of NHB patients in the 73 transplant centers.

compositions across centers underscored the need to investigate the potential impact of site of care on disparities in graft failure times after kidney transplantation for different racial groups.

To answer the question of whether there exist site-of-care-associated racial disparities, we fitted the dataset using the proposed CEDAR-t2e framework. Specifically, we consider $K = 3$ knots for the spline basis in each site, and set t_{j1}, t_{j2}, t_{j3} as the 40%, 60%, and 80% sample quantiles for the observed graft failure times in site j. In site-level meta-regression in Module I, given the limited site-level covariate availability, we only consider one site-level covariate, that is, the total number of patients in the site, to represent the volume of the transplant center. Based on the above setting, we calculated the difference between the counterfactual RMST and the factual RMST of NHB patients within 3 windows, 4-month, 8-month, and 1-year windows, where "counterfactual" means that we assumed NHB patients were admitted to transplant centers in the same distribution as NHW patients were admitted. For each one of the 3 times, we repeated the simulation procedure 400 times to account for uncertainties in the simulation. As a result, there were 400 values for the difference between the counterfactual and factual RMSTs of NHB patients for each window.

The empirical distribution of the difference between the counterfactual and factual RMSTs for different time windows is summarized in Figure 4. The results revealed the statistically significant though very small site-of-care-associated racial disparities in graft failure times after kidney transplantation: the average difference between counterfactual and factual RMSTs of NHB patients within 4-month, 8-month, and 1-year periods were 0.120 (95% CI, 0.109-0.132), 0.337 (95% CI, 0.307-0.372), and 0.613 (95% CI, 0.559-0.675) days, respectively. If NHB patients had the same access to transplant centers as NHW patients, their lifespan would extend about 0.12, 0.34, and 0.61 days when confined in the 4-month, 8-month, and 1-year period, respectively. The magnitude of the extension in graft failure times is almost negligible compared with the length of the time windows, which indicated that site of care was not a clinically significant contributor to racial disparities in post-transplantation survival times.

Discussion

This study investigated if the site of care is a significant contributor to racial disparities in post-transplantation survival times between NHB and NHW patients. Unlike most existing studies with a focus on racial disparities in mortality rates, we examined time-to-event outcomes which are more informative and provide granular quantification of patients' lifespans. To address the data-sharing barrier arising from regulatory concerns, we developed a federated learning framework called CEDAR-t2e, which only required sharing summary-level statistics across hospitals. To quantitatively account for site-of-care associated racial disparities in post-transplantation survival times, we utilized the counterfactual modeling approach to calculate how long the graft failure time of NHB patients would be extended had they been admitted to transplant centers in the same distribution as NHW patients were admitted. This value is an intuitive indicator showing whether the site of care is a main contributor to racial disparities. With the application of the proposed method to a real registry dataset from 73 transplant centers, we found the expected post-transplantation survival time for NHB patients during the one year after transplantation would be only 0.61 days longer if they had the same access to transplant centers as NHW patients. The findings from our dataset analysis suggest no meaningful site-of-care related racial disparities in survival times following kidney transplantation.

The ability to rigorously assess site-of-care-associated disparities is a critical first step toward advancing health equity and improving patient outcomes. Identifying whether site-of-

Downloaded from https://academic.oup.com/jamia/article/32/12/1916/8262716 by guest on 05 March 2026

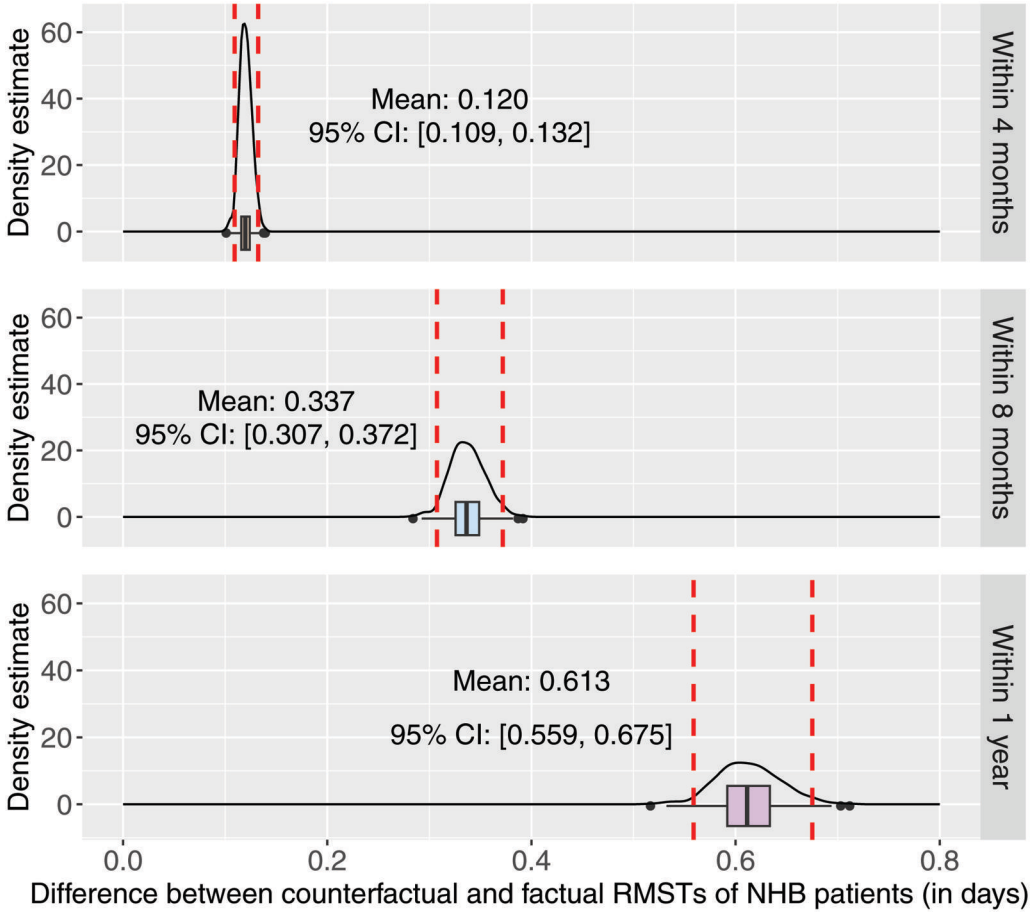


Figure 4. The difference between counterfactual RMSTs and factual RMSTs of NHB patients within 4-month, 8-month, and 1-year windows, assuming that NHB patients were admitted to transplant centers in the same distribution as NHW patients were admitted. The vertical red lines indicate the 95% CIs.

care factors meaningfully contribute to racial disparities enables healthcare systems to target interventions appropriately. When site-of-care disparities are detected, systems can implement hospital-level quality improvement efforts, allocate resources to under-resourced centers, or standardize care processes to promote equitable access to high-quality care.

Conversely, when site-of-care disparities are not detected—as demonstrated in our study—this finding is equally informative. It suggests that hospital-level interventions may have limited impact in this setting, and efforts should instead prioritize addressing upstream factors such as access to transplantation, pre-transplant care, social determinants of health, or structural inequities.

To the best of our knowledge, this study is the first to employ a federated learning algorithm in evaluating the site-of-care associated racial disparities in time-to-event outcomes. The proposed approach has several advantages. First, to facilitate federating learning of the model for time-to-event outcomes, we leveraged the technique of Poissonization to transform the proportional hazards model to an equivalent GLMM. This greatly simplified model fitting and parameter estimation, making the estimation procedures in each local hospital much easier to implement. Second, it is communication-efficient in that it requires only 3 rounds of communication between each hospital and the lead hospital. Each round of between-hospital communication only involves transferring some parameter estimates that consume very little memory, and thus the communication can be completed efficiently at low memory costs. In practice, the total time required for the 3 rounds of communication is typically on the order of a few days, primarily due to the involvement of human coordination during each synchronization step. The actual computational time for each round is relatively minimal—ranging from a few seconds to a few minutes within each local site, depending on the sample size and computational resources. Third, the proposed CEDAR-t2e framework avoids sharing individual-level information across hospitals. It also avoids sharing sensitive information at the hospital level. More specifically, in the counterfactual modeling module, each hospital locally calculates the counterfactual/factual RMSTs for their patients. The lead hospital has access to the results of counterfactual/factual RMSTs of all participating hospitals, while each local hospital only has access to its own counterfactual/factual RMSTs.

Within this study, we also find a few promising directions for future research. From the improvement of statistical method perspective, first, it is possible that there exist some unmeasured factors that may confound the site-of-care related disparities. It would be helpful if we could verify whether there exists unmeasured confounding with the help of negative control outcomes.[29] Second, the current method implicitly assumes that the effect of patient-level covariates on the event time is time-invariant, while this assumption may not hold in practice. For example, conditions like

Downloaded from https://academic.oup.com/jamia/article/32/12/1916/8262716 by guest on 05 March 2026

diabetes might have an increasing effect on graft failure risk as it progresses or worsen. It would be interesting and lead to a more accurate assessment of the site-of-care related disparities if we could incorporate such time-varying effects into our current federated learning framework. Third, the current method cannot handle missingness in patient-level covariates. We plan to address this challenge by incorporating some statistical imputation techniques[30] into the current framework. Fourth, the current model and the Poissonization technique require the assumption that the occurrence of failures for different patients are independent. This assumption means that there should not be any clustering effect among patients. In practice, there might be some scenarios that this assumption does not hold. In this case, one potential way to incorporate the dependence between patients is to introduce a common latent variable for the group of patients with dependence. This would lead to additional challenges for statistical estimation and subsequent counterfactual modelling, which is a promising direction for future research. Fifth, the current approach assumes that all participating site are willing to share necessary summary-level statistics. It is also important to consider potential worst-case privacy risks, especially in situations where certain hospitals may be unwilling or unable to share any information related to race or ethnicity. In such cases, one possible approach is to exclude these hospitals from the analysis, though this may result in a loss of valuable information and limit the generalizability of the findings. An alternative and promising direction is the incorporation of homomorphic encryption[39] (HE) into the federated learning framework. Homomorphic encryption allows computations to be performed directly on encrypted data without requiring decryption. In the context of CEDAR-t2e, hospitals could use HE to encrypt their summary statistics or model updates before transmission. The coordinating center would then be able to aggregate these encrypted values and carry out necessary computations while preserving data confidentiality. This approach enhances protection against privacy breaches and may enable broader participation from institutions with stringent data-sharing policies. However, implementing HE in federated learning remains computationally intensive and requires careful algorithmic design to ensure scalability and accuracy. Addressing these technical challenges is an important avenue for future work. From a RWD analysis perspective, it is of interest to apply the proposed framework to larger datasets that include more diverse racial categorizations and examine potential site-of-care-associated racial disparities across different races. It is important to note that the proposed method is broadly applicable across multiple datasets and a variety of disease areas beyond kidney transplantation. Its aim is to address various health inequities across diverse populations and healthcare settings.

Conclusions

We proposed the CEDAR-t2e framework to answer whether there exist site-of-care associated racial disparities in time-to-event outcomes. The proposed method leveraged the equivalent transformation from proportional hazards models to GLMMs. It is a communication-efficient algorithm with only 3 rounds of communication in summary-level statistics. We believe that CEDAR-t2e is a significant contribution to this new generation of DRNs. Based on our analysis of a decentralized time-to-event dataset covering 73 kidney transplant centers, we found the site of care did not contribute to the racial disparities in post-kidney-transplantation survival times.

Author contributions

Yudong Wang (Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing—original draft, Writing—review & editing), Dazheng Zhang (Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing—original draft, Writing—review & editing), Jiayi Tong (Conceptualization, Data curation, Formal analysis, Methodology, Visualization, Writing—review & editing), Xing He (Data curation, Formal analysis), Liang Li (Methodology), Lichao Sun (Data curation), Ashutosh M. Shukla (Data curation), Jiang Bian (Conceptualization, Data curation, Supervision), David A. Asch (Conceptualization, Methodology, Supervision, Writing—review & editing), and Yong Chen (Conceptualization, Data curation, Formal analysis, Funding acquisition, Methodology, Supervision, Writing—review & editing)

Supplementary material

Supplementary material is available at *Journal of the American Medical Informatics Association* online.

Funding

Yudong Wang, Dazheng Zhang, Jiayi Tong, and Yong Chen were supported in part by National Institutes of Health (U01TR003709, U24MH136069, RF1AG077820, R01AG073435, R56AG074604, R01LM013519, 1R01LM014344, R01DK128237, R21AI167418, and R21EY034179). Lichao Sun was partially supported by the National Science Foundation Grants (CRII-2246067, ATD-2427915, NSF POSE-2346158, POSE-2449280) and Lehigh Grant (FRGS00011497).

Conflicts of interest

The data reported here have been supplied by the USRDS. The interpretation and reporting of these data are the responsibility of the author(s) and in no way should be seen as an official policy or interpretation of the U.S. government.

Data availability

While the deidentified aggregate results are disclosed within this publication, the data cannot be released to the public or the journal. Should verification of findings be necessary, the journal or any interested entities may enter into a Data Use Agreement (DUA) directly with the USRDS. The R code is available at https://github.com/Penncil/CEDAR-t2e. The analyses were conducted using R version 4.2.1.

References

1. Hariharan S, Israni AK, Danovitch G. Long-term survival after kidney transplantation. *N Engl J Med*. 2021;385:729-743.
2. Garcia GG, Harden P, Chapman J, World Kidney Day Steering Committee 2012. The global role of kidney transplantation. *Lancet*. 2012;379:e36-e38.

Downloaded from https://academic.oup.com/jamia/article/32/12/1916/8262716 by guest on 05 March 2026

3. Tonelli M, Wiebe N, Knoll G, et al. Systematic review: kidney transplantation compared with dialysis in clinically relevant outcomes. *Am J Transplant*. 2011;11:2093-2109.
4. Gill JS, Lan J, Dong J, et al. The survival benefit of kidney transplantation in obese patients. *Am J Transplant*. 2013;13:2083-2090.
5. Heldal K, Hartmann A, Grootendorst DC, et al. Benefit of kidney transplantation beyond 70 years of age. *Nephrol Dial Transplant*. 2010;25:1680-1687.
6. Asch DA, Islam MN, Sheils NE, et al. Patient and hospital factors associated with differences in mortality rates among Black and White US medicare beneficiaries hospitalized with COVID-19 infection. *JAMA Netw Open*. 2021;4:e2112842.
7. Kulkarni S, Ladin K, Haakinson D, et al. Association of racial disparities with access to kidney transplant after the implementation of the new kidney allocation system. *JAMA Surg*. 2019;154:618-625.
8. Malek SK, Keys BJ, Kumar S, Milford E, Tullius SG. Racial and ethnic disparities in kidney transplantation. *Transpl Int*. 2011;24:419-424.
9. Liu A, Woodside KJ, Augustine JJ, Sarabu N. Racial disparity in kidney transplant survival relates to late rejection and is independent of steroid withdrawal. *Clin Transplant*. 2018;32:e13381.
10. Patzer RE, Mohan S, Kutner N, McClellan WM, Amaral S. Racial and ethnic disparities in pediatric renal allograft survival in the United States. *Kidney Int*. 2015;87:584-592.
11. Taber DJ, Gebregziabher M, Srinivas T, Egede LE, Baliga PK. Transplant center variability in disparities for African-American kidney transplant recipients. *Ann Transplant*. 2018;23:119-128.
12. Chen Y, Li Y, Liu Y, et al. Residential racial and ethnic segregation and post-kidney transplant dementia, allograft loss, and mortality. *Am J Kidney Dis*. 2024;84:258-261.
13. Li Y, Menon G, Kim B, et al. Neighborhood segregation and access to live donor kidney transplantation. *JAMA Intern Med*. 2024;184:402-413.
14. Asch DA, Sheils NE, Islam MN, et al. Variation in US hospital mortality rates for patients admitted with COVID-19 during the first 6 months of the pandemic. *JAMA Intern Med*. 2021;181:471.
15. Hripcsak G, Duke JD, Shah NH, et al. Observational Health Data Sciences and Informatics (OHDSI): opportunities for observational researchers. *Stud Health Technol Inform*. 2015;216:574-578.
16. STAR Clinical Research Network. Accessed April 26, 2024. https://starcrn.org/
17. Brat GA, Weber GM, Gehlenborg N, et al. International electronic health record-derived COVID-19 clinical course profiles: the 4CE consortium. *NPJ Digit Med*. 2020;3:109.
18. Tong J, Reps JM, Luo C, et al. Unlocking efficiency in real-world collaborative studies: a multi-site international study with one-shot lossless GLMM algorithm. *Npj Digit Med*. 2025;8:457.
19. Duan R, Luo C, Schuemie MJ, et al. Learning from local to global: An efficient distributed algorithm for modeling time-to-event data. *J Am Med Inform Assoc*. 2020;27:1028-1036.
20. Wu Q, Reps JM, Li L, et al. COLA-GLM: collaborative one-shot and lossless algorithms of generalized linear models for decentralized observational healthcare data. *Npj Digit Med*. 2025;8:442.
21. Case-mix measurement and assessing quality of hospital care. *Health Care Financ Rev*. 1987;1987:39-48.
22. Wu Y, Jiang X, Kim J, Ohno-Machado L. Grid Binary LOgistic REgression (GLORE): building shared models without sharing data. *J Am Med Inform Assoc*. 2012;19:758-764.
23. Luo C, Islam MN, Sheils NE, et al. dPQL: a lossless distributed algorithm for generalized linear mixed model with application to privacy-preserving hospital profiling. *J Am Med Inform Assoc*. 2022;29:1366-1371.
24. Yan Z, Zachrison KS, Schwamm LH, Estrada JJ, Duan R. A privacy-preserving and computation-efficient federated algorithm for generalized linear mixed models to analyze correlated electronic health records data. *PLoS One*. 2023;18:e0280192.
25. PDA-OTA. Accessed April 26, 2024. https://pda-ota.pdamethods.org/login
26. Annual Data Report | USRDS. Accessed April 26, 2024. https://usrds-adr.niddk.nih.gov/2023
27. Luo C, Duan R, Naj AC, et al. ODACH: a one-shot distributed algorithm for Cox model with heterogeneous multi-center data. *Sci Rep*. 2022;12:6627.
28. Zhang D, Tong J, Stein R, et al. One-shot distributed algorithms for addressing heterogeneity in competing risks data across clinical sites. *J Biomed Inform*. 2024;150:104595.
29. Ding Y, Nan B. A sieve M-theorem for bundled parameters in semiparametric models, with application to the efficient estimation in a linear model for censored data. *Ann Stat*. 2011;39:2795-3443.
30. Zhao X, Wu Y, Yin G. Sieve maximum likelihood estimation for a general class of accelerated hazards models with bundled parameters. *Bernoulli*. 2017;23:3385-3411.
31. Li L, Hu B, Greene T. A semiparametric joint model for longitudinal and survival data with application to hemodialysis study. *Biometrics*. 2009;65:737-745.
32. Cai T, Hyndman RJ, Wand MP. Mixed model-based hazard estimation. *J Computat Graph Stat*. 2002;11:784-798.
33. Liang CJ, Luo C, Kranzler HR, Bian J, Chen Y. Communication-efficient federated learning of temporal effects on opioid use disorder with data from distributed research networks. *J Am Med Inform Assoc*. 2025;32:656-664.
34. van Houwelingen HC, Arends LR, Stijnen T. Advanced methods in meta-analysis: multivariate approach and meta-regression. *Stat Med*. 2002;21:589-624.
35. Efron B, Morris C. Data analysis using Stein's estimator and its generalizations. *J Am Stat Assoc*. 1975;70:311-319.
36. Royston P, Parmar MKB. The use of restricted mean survival time to estimate the treatment effect in randomized clinical trials when the proportional hazards assumption is in doubt. *Stat Med*. 2011;30:2409-2421.
37. Royston P, Parmar MKB. Restricted mean survival time: an alternative to the hazard ratio for the design and analysis of randomized trials with a time-to-event outcome. *BMC Med Res Methodol*. 2013;13:152.
38. Zhao L, Claggett B, Tian L, et al. On the restricted mean survival time curve in survival analysis. *Biometrics*. 2016;72:215-221.
39. Acar A, Aksu H, Uluagac AS, Conti M. A survey on homomorphic encryption schemes: theory and implementation. *ACM Comput. Surv*. 2018;51:1–35.

Downloaded from https://academic.oup.com/jamia/article/32/12/1916/8262716 by guest on 05 March 2026

© The Author(s) 2025. Published by Oxford University Press on behalf of the American Medical Informatics Association. All rights reserved.
For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site—for further information please contact journals.permissions@oup.com.
Journal of the American Medical Informatics Association, 2025, 32, 1916–1926
https://doi.org/10.1093/jamia/ocaf138
Research and Applications